

CT AND MRI FEATURES OF CAROTID BODY PARAGANGLIOMAS IN 16 DOGS

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Carotid body tumors (paragangliomas) arise from chemoreceptors located at the carotid bifurcation. In imaging studies, this neoplasm may be confused with other neck neoplasms such as thyroid carcinoma. The purpose of this retrospective, cross-sectional study was to describe computed tomographic (CT) and magnetic resonance imaging (MRI) characteristics of confirmed carotid body tumors in a multi-institutional sample of dogs. A total of 16 dogs met inclusion criteria (14 examined using CT and two with MRI). The most common reason for imaging was a palpable cervical mass or respiratory signs (i.e., dyspnea or increased respiratory noises). The most commonly affected breed was Boston terrier ($n = 5$). Dogs were predominantly male castrated ($n = 10$) and the median age was 9 years [range 3–14.5]. Most tumors appeared as a large mass centered at the carotid bifurcation, with poor margination in six dogs and discrete margins in ten dogs. Masses were iso- to hypodense to adjacent muscles in CT images and hyperintense to muscles in T1- and T2-weighted MRI. For both CT and MRI, masses typically showed strong and heterogeneous contrast enhancement. There was invasion into the adjacent structures in 9/16 dogs. In six of these nine dogs, the basilar portion of the skull was affected. The external carotid artery was entrapped in seven dogs. There was invasion into the internal jugular vein in three dogs, and into the external jugular, maxillary, and linguo-facial veins in one dog. Imaging characteristics helped explain some clinical presentations such as breathing difficulties, Horner's syndrome, head tilt, or facial nerve paralysis. © 2015 American College of Veterinary Radiology.

Key words: carotid body tumor, chemodectoma, CT, dog, MRI, paraganglioma.

Introduction

CAROTID BODY TUMORS (paragangliomas, chemodectomas, *glomus* tumors) are an uncommon cause of neck masses in dogs.^{1–4} They arise from paraganglionic cells that embryologically originated from the neural crest and migrated along the branchial arches, in close association with autonomic ganglion cells.¹ In the adult, extra-adrenal paraganglia form clusters of neuroendocrine cells that are closely associated with the autonomous nervous system.² They are present at several sites including the carotid bodies, aortic bodies, nodose ganglion of the vagus nerve, ciliary ganglion in the orbit, pancreas, bodies on the internal jugular vein below the middle ear and *glomus jugulare* along the recurrent branch of the glossopharyngeal nerve.³ In people,

head and neck paragangliomas are located at four typical locations in the carotid space and skull base region.⁵ The type of paraganglioma is determined based on the paraganglion of origin, which is determined from imaging, surgical, or pathologic findings.^{6–8} When paragangliomas are located in the carotid bifurcation, they are called carotid body tumors. Vagal paragangliomas typically arise from nests of paraganglionic tissue, within the perineurium of the vagal nerve; this includes the nodose ganglion below the skull base within the carotid fascial space, and the superior ganglion of the vagus nerve in the jugular fossa. Tympanic paragangliomas arise along the tympanic branch of the glossopharyngeal nerve, and can be found anywhere along the course of this nerve. Jugular paragangliomas originate within the adventitia of the jugular bulb. Both jugular and tympanic paragangliomas cause symptoms by physically expanding and compressing surrounding structures, especially the accompanying nerves. Because of the proximity of their structures of origin, larger tympanic, or jugular paragangliomas can often not be differentiated, and are therefore called jugulo-tympanic paragangliomas.^{6–8}

In dogs, chemodectomas develop mainly in the aortic bodies, close to the heart base, and less commonly in the carotid bodies, located at the level of the bifurcation of the

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common carotid arteries into internal and external carotid arteries.³ Other than the carotid bodies, there are only two canine case reports of a noncarotid head and neck paraganglioma, one *glomus jugulare* and one jugulo-tympanic paraganglioma invading the tympanic bulla and external ear canal.^{1,9} Because they are usually slowly growing and nonsecreting, carotid body tumors are often diagnosed late, when enlarging tumors interfere with swallowing, cause dyspnea, or cause circulatory disturbances from compression of the large veins of the neck.^{3,10–19} Although these tumors are most commonly benign, metastases occur in about 30% of dogs.^{3,20} Sites of metastasis include lungs, regional lymph nodes, liver, pancreas, and kidneys.⁴ Metastases to the mediastinum and bones have also been reported, with an apparent predisposition for the vertebrae.^{1,16,20,21} Clinico-pathological features of canine carotid body tumors have been extensively described in the literature.^{10–26} Diagnosis has primarily been based on the presence of a mass at the carotid bifurcation that was detected using radiographs, ultrasound, CT, or MRI. It has been reported, and it has been our experience, that this neoplasm may be confused with other tumors such as thyroid carcinomas.²⁵ This confusion is probably in part due to a lack of awareness of the imaging features of carotid body tumors, as well as their neuroendocrine nature that can make cytologic diagnosis difficult. To our knowledge, the CT and MRI features of carotid body tumors have only been reported in two case reports,^{11,14} and briefly illustrated in two case series, which were not focused on the imaging features of these tumors.^{17,25}

The purpose of this retrospective, cross-sectional study was to describe the CT and MRI features of confirmed paragangliomas of the carotid region in a multi-institutional sample group of dogs, and to determine whether these characteristics are similar to those described in people.

Materials and Methods

Medical record databases of three university veterinary teaching hospitals (University of Pennsylvania ($n = 7$), North Carolina State University ($n = 5$), and University of California, Davis ($n = 4$)) were searched for dogs with a confirmed diagnosis of carotid body tumors that also received a CT or MRI examination of the head and neck area at the time of diagnosis. Diagnosis had to be confirmed within 2 weeks of the imaging study, and confirmation was through cytology, histopathology of a biopsy or surgical sample, or necropsy, together with an anatomic location consistent with the carotid bodies (carotid bifurcation). Data obtained from the medical records included dogs' age, breed, sex, and clinical reason for imaging. Reports from cytopathology, histopathology, surgery, or necropsy were also recorded.

All imaging studies included in this series were uploaded to a DICOM-viewing station (Philips iSite Radiology 3.5.0. Royal Philips Electronics, Amsterdam, The Netherlands). A board-certified veterinary radiologist (W.M.) reviewed all images available. The radiologist also used CT multi-planar (MPR) and oblique MPR reformatting software tools to characterize masses. The initial radiology report was also reviewed and initial impressions of the radiologist at the time of the study were recorded. Imaging characteristics recorded were location of the tumor, size (length, height, and width), nature of mass effect on the carotid arteries, presence of entrapment of the external carotid artery (defined as enhancing tissue surrounding the carotid artery on postcontrast series), shape (rounded, oval, irregular), margination (well or poorly marginated, with well-defined margins defined as the presence of a distinct border to surrounding tissues on the postcontrast series), signal characteristics (in comparison to adjacent muscles), enhancement pattern, presence or absence, and nature of local invasion, presence or absence of loco-regional lymphadenopathy (enlargement was subjectively assessed for the mandibular lymph nodes while enlargement of the medial retropharyngeal lymph node was defined as >2 cm width or >1 cm height).²⁷

Results

Sixteen dogs met the inclusion criteria, 14 of which received a CT and two of which received an MRI. The study period spanned 11 years. Breeds included were five Boston terriers, and one each of Labrador, Rottweiler, Shi-Tzu, Collie, English bulldog, Boxer, Corgi, Bernese mountain dog, American Pit bull terrier, Cavalier King Charles Spaniel, and French bulldog. Six were female spayed and 10 were male castrated. Median age was 9 years (range 3–14.5). All CT studies included a transverse series over the head and neck in a soft tissue and bone algorithm and a postcontrast series in a soft tissue algorithm. Depending on the time frame and imaging center, contrast material injection had been performed manually or using a power injector. All MRI studies included at least a transverse T1- and T2-weighted series and transverse T1-W series after manual injection of gadolinium. Specific technical parameters for CT and MRI studies varied widely.

The reason for CT or MRI imaging included a mass in the cervical region that was palpated by the owner ($n = 3$) or found during intubation for neutering ($n = 1$) or dental prophylaxis ($n = 1$). Increased respiratory noise, stertor, and/or dyspnea were present in seven dogs. Horner's syndrome was present in three dogs. In one of these three dogs, swallowing difficulties were also present, and in the other two dogs, other cranial nerves deficits were present including facial nerve paralysis in two dogs, and hypoglossal nerve deficit in one dog. One of these dogs had a head tilt, and

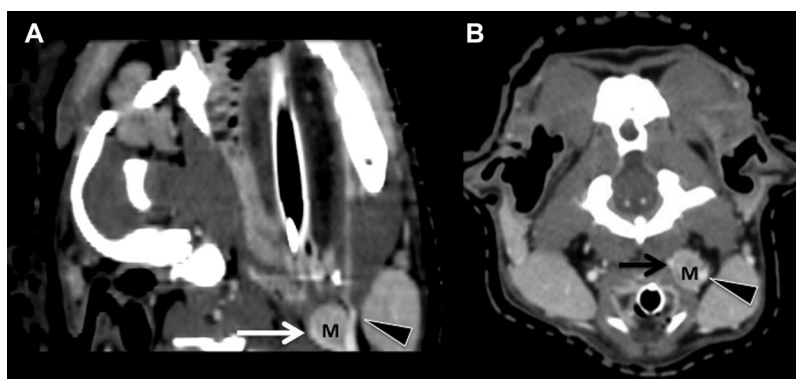


FIG. 1. Postcontrast CT images in a dog showing a small carotid body tumor on the left side. (A) Dorsal oblique reformatted image in a soft tissue window. (B) Transverse image at the level of the mass (M). The mass (M) is seen at the carotid bifurcation, splaying the external carotid artery (arrowhead) and the internal carotid artery (arrow).

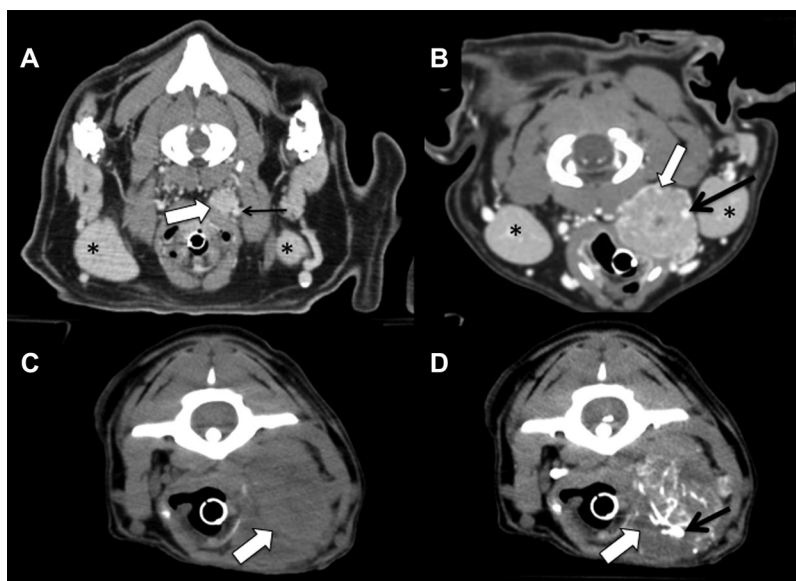


FIG. 2. (A) Postcontrast transverse CT image in a dog showing a small, well marginated, strongly enhancing carotid mass; (B) Postcontrast transverse CT image in a dog showing a large, well marginated, strongly enhancing carotid mass; (C and D) Precontrast (C) and postcontrast (D) CT images of a large, poorly marginated carotid body mass. For all images the following legend applies: Black arrow = external carotid artery; white arrow: carotid body tumor; * = mandibular salivary glands.

a mass was identified in the ear canal during otoscopic inspection. Biopsy identified a neuroendocrine tumor, and the dog was referred for CT and radiation planning. In the two dogs with an MRI examination, the carotid body mass was an incidental finding during imaging of the brain or cervical spine for neurologic signs. An additional dog received head and neck CT for radiation planning after a mass in the carotid area had been identified on the edge of the field of view of an MRI for a brain tumor (temporal lobe glioma); only the CT study was included in this case series. Finally, a dog with a history of previous histiocytic sarcoma received a whole body CT for cancer screening after developing biventricular effusion, and the carotid body tumor was found incidentally at that time. Histopathological examination in 13 dogs revealed a neuroendocrine tumor consistent with a

carotid body tumor. Immunohistochemistry was only performed in one dog and confirmed staining consistent with a paraganglioma. In three dogs, cytological examination revealed cells with characteristics consistent with a neuroendocrine tumor. These characteristics in addition to the location of the lesions as seen on cross-sectional imaging lead to a clinico-pathological diagnosis of carotid body tumor.

As this was one of the inclusion criteria, CT or MRI examination in all dogs revealed a mass at or extending cranial to the carotid bifurcation. Splaying of the carotid bifurcation could be identified in two dogs with relatively small masses, with lateral displacement of the external carotid artery and medial displacement of the internal carotid artery (Fig. 1). In most dogs however, the internal carotid

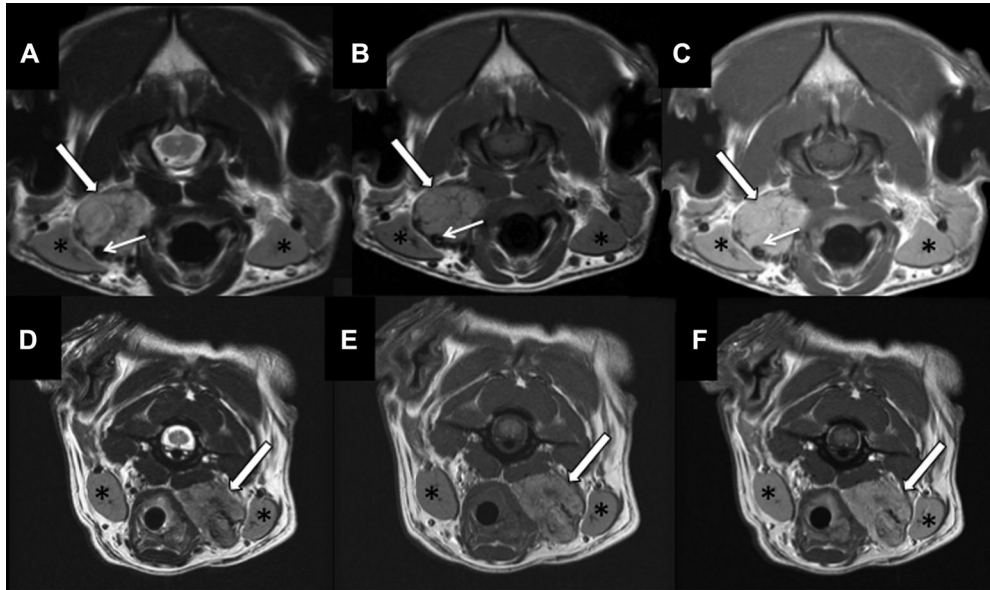


FIG. 3. Each row corresponds to MRI images at the level of a carotid body tumor on the right side (A, B, C) and on the left side (D, E, F) in two different dogs. Images are T2-weighted (A, D), T1-weighted precontrast (B, E), and T1-weighted postcontrast (C, F). The following legend applies to all images; large white arrow = carotid body tumor; small white arrow: external carotid artery; * = mandibular salivary gland. The carotid body tumors are hyperintense to the muscles on all series and strongly contrast enhancing.

artery could not be identified anymore as this tiny vessel was obliterated by the larger mass. The contralateral carotid bifurcation was utilized to localize the mass in relation to the level of the carotid bifurcation in these dogs. In one dog there was no displacement of the ipsilateral external carotid artery, but in all other dogs there was some degree of lateral or ventrolateral displacement of the external carotid artery, with more displacement seen with larger masses. The external carotid artery was entrapped by neoplastic tissue in seven dogs. Normal thyroid glands were seen in all dogs in their normal location, caudal to the neoplastic mass. They could be recognized on precontrast CT series due to their characteristic hyperattenuation.

The paraganglioma was on the left side in nine dogs and on the right side in seven dogs. Median length was 4.75 cm (range 1.00–9.00), median width was 3.05 cm (range 1.30–6.10), and the median height was 3.10 cm (range 0.98–5.30). The masses were oval-shaped in six dogs, spherical in five dogs and irregularly shaped in five dogs. They were poorly margined in six dogs and well margined in ten dogs (Fig. 2).

On precontrast CT images ($n = 14$), the mass was homogeneous and isoattenuating to muscles in six dogs, homogeneous and mildly hypoattenuating to muscles in five dogs, and heterogeneous with multifocal hypoattenuating areas in three dogs. In two dogs, small speckles of mineralization were present within the masses. After iodinated contrast material injection, strong and heterogeneous enhancement was present in 12 dogs. Moderate contrast enhancement was present in two dogs, with a rim-like pattern in one of these two dogs.

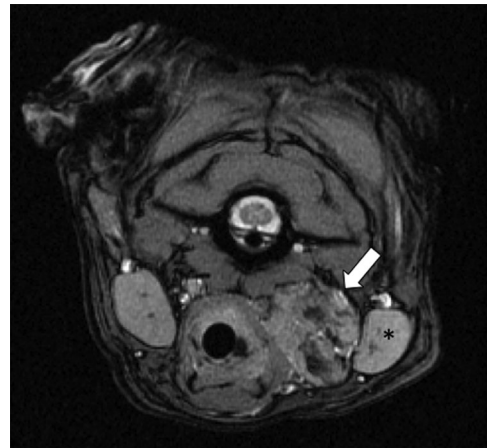


FIG. 4. Gradient-echo transverse image of a left-sided carotid body tumor (white arrow) in a dog showing numerous hypointense foci consistent with hemorrhage. * = mandibular salivary gland.

In the two dogs with MRI, the masses were hyperintense to muscles on both T1- and T2-weighted images (Fig. 3). In one of the two dogs a gradient echo pulse sequence was available, showing multiple hypoattenuating foci suggestive of internal hemorrhage (Fig. 4). Both masses strongly and heterogeneously enhanced after gadolinium injection, with degree of enhancement equal or slightly more than that of the mandibular salivary glands (Fig. 3).

There was some form of invasion into the adjacent structures in 9/16 dogs. In six of these nine dogs, the basilar portion of the skull was affected including invasion and widening of the ipsilateral petro-occipital fissure in four

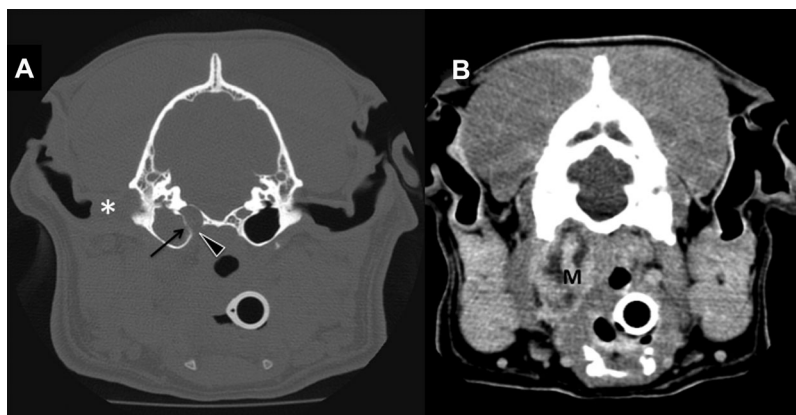


FIG. 5. Invasive carotid body tumor in a dog on the right side. (A) Precontrast CT image in a bone window at the level of the tympanic bullae showing widening of the right petro-occipital fissure due to tumor extension (arrowhead), thinning, and lysis of the medial wall of the right tympanic bulla (small black arrow) with neoplastic tissue extending into the right tympanic bulla and right external ear canal (*). (B) Postcontrast CT image in the same dog in a soft tissue window caudal to the tympanic bullae showing the poorly margined, strongly and heterogeneously contrast-enhancing carotid body mass (M).

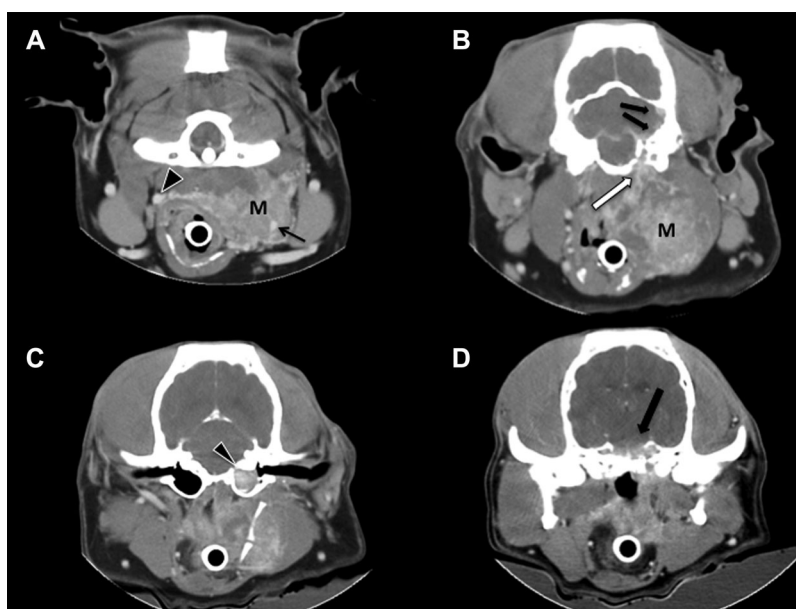


FIG. 6. Postcontrast, serial transverse images (A to D: caudal to rostral) in a dog with a left sided carotid body tumor. (A) The heterogeneously enhancing mass is visible (M) displacing the left external carotid artery (small black arrow) latero ventrally. The black arrowhead indicates the right carotid artery. (B) Extension of the enhancing neoplastic tissue is seen toward the left petro-occipital fissure, which is widened (white arrow). Enhancing tissue is seen along the left ventro-lateral margin of the cerebellum (black arrows). (C) The left tympanic bulla (black arrowhead) is filled with enhancing neoplastic tissue. (D) Enhancing tissue is seen along the ventral margin of the cranial cavity (black arrow).

dogs (Figs. 5, 6, and 7); lysis of the medial wall of the ipsilateral tympanic bulla without gross invasion into the lumen of the bulla in three dogs (Fig. 7); invasion into the ipsilateral bulla in two dogs (Figs. 5, 6), with extension into the external ear canal in one of these two dogs (Fig. 5); lysis of the basisphenoid bone in one dog; extension into the brain cavity, with visible enhancing neoplastic tissue causing mass effect onto the cerebellum adjacent to the basisphenoid and petrous temporal bones in three dogs (Figs. 6, 7). In these three dogs, invasion of the caudal fossa occurred through the petro-occipital fissure, entering

the brain cavity through the jugular foramen. In one of these three dogs, additional dorsal extension ventral to the brain occurred through lysis of the rostro-dorsal margin of the tympanic bulla that was invaded by neoplastic tissue (Fig. 6); in another one of these three dogs, neoplastic tissue was seen extending caudally through the foramen magnum in the cranial cervical spine to the level of the atlas, causing left-sided cord compression. Necropsy in one of these dogs with cranial cavity invasion revealed that the meninges on the left lateral aspect of the medulla oblongata and cerebellum were expanded and filled by the same neoplastic tissue



FIG. 7. CT images in a dog with a large, poorly margined, heterogeneously contrast-enhancing mass in the left carotid body; (A) Dorsal oblique maximum intensity projection (MIP), (B) transverse postcontrast image, in a soft tissue window, and (C) transverse precontrast image in a bone window. On the dorsal MIP (A), the mass (M) is seen to extend cranial to the carotid body bifurcation and displaces the left external carotid artery laterally (LECA). The right carotid bifurcation is indicated by the black arrow; the asterisks (*) indicates the mandibular salivary glands. In (B), one can see extension of enhancing neoplastic tissue into the brain cavity along the left ventral aspect of the cerebellum (black arrowhead). In (C), widening of the left petro-occipital fissure and lysis of the medial wall of the left tympanic bulla is noted (white arrow).

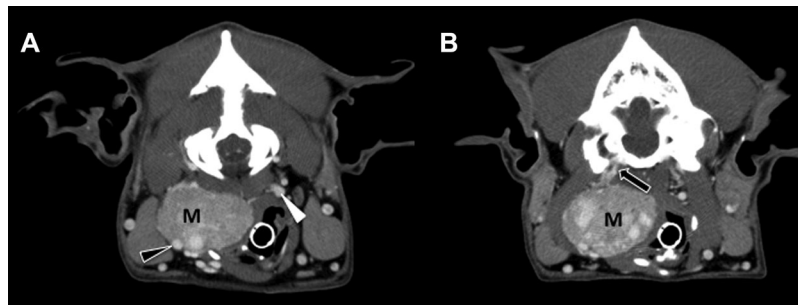


FIG. 8. Postcontrast CT images in a dog with a right-sided large carotid body paraganglioma (M). In A, ventro-lateral displacement of the right external carotid artery is seen (black arrowhead) compared to the left carotid artery (white arrowhead). In B, the mass (M) is seen to send a stalk of enhancing tissue toward the right petro-occipital fissure.

as that identified at the carotid bifurcation. In three additional dogs, although there was no invasion into the skull base yet, a stalk of enhancing neoplastic tissue was visible on postcontrast images extending toward and very close to the ipsilateral petro-occipital fissure (Fig. 8).

Other than the carotid artery entrapment-mentioned above, vascular invasion was noted in four dogs. There was invasion into the internal jugular vein in three dogs, and invasion of the ipsilateral external jugular vein, maxillary, and linguo-facial veins in one dog (Fig. 9). There was lysis of the ipsilateral portion of the thyroid cartilage in one dog and mild lysis of the adjacent hyoid bones in one other dog. Invasion into the surrounding muscles was present in one dog and effacement of the ipsilateral medial retropharyngeal lymph node was noted in four dogs. Two of the three

dogs with brain cavity invasion had a Horner's syndrome with left hypoglossal nerve paresis in one and swallowing disorder in the other. The two dogs with neoplastic tissue within the tympanic bulla had a Horner's syndrome and concurrent facial paralysis; one of these two dogs had concurrent brain cavity invasion.

In 5/16 dogs, no significant satellite lymphadenopathy was noted on the imaging study. Histopathology was performed on the ipsilateral medial retropharyngeal lymph node in one of these five dogs (removed with the mass at surgery), and revealed a small metastatic nodule. No histopathological or cytological examinations of the nodes were performed in the four other dogs. In 4/16 dogs, the medial retropharyngeal lymph node ipsilateral to the carotid body tumor was effaced by the mass and not identified

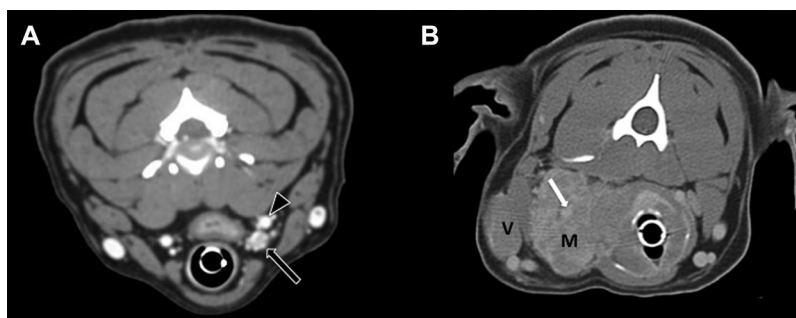


FIG. 9. Transverse postcontrast images in two different dogs with carotid body tumors that had venous invasion. In A, the internal jugular vein (black arrow) is dilated and filled with heterogeneous material, consistent with tumor thrombosis; the black arrowhead indicates the left carotid artery. In B, the maxillary vein (V) is markedly distended and filled with heterogeneous soft tissue material; the carotid body mass is seen (M) and has encased the right external carotid artery (white arrow).

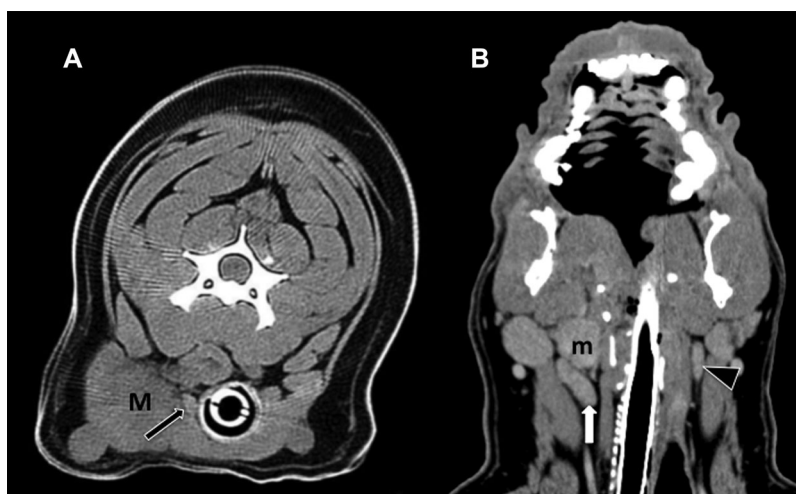


FIG. 10. Two examples where the carotid body tumor was misinterpreted initially for a thyroid tumor (A) and a neoplastic medial retropharyngeal lymph node (B). (A) Transverse CT image before contrast administration in a dog with a right-sided carotid body paraganglioma; the caudal edge of the mass is visible (M) and the right thyroid gland was visible medial to the mass, appearing hyperattenuating due to its iodine content. (B) Dorsal reconstruction of a postcontrast CT examination in a dog. The right medial retropharyngeal lymph node (white arrow) is seen displaced caudally and laterally by the carotid body mass (m). The left medial retropharyngeal lymph node is visible in a normal location (black arrowhead).

on the CT images. Histopathology of the entire mass removed at surgery was performed in only 1/4 dogs and confirmed lymph node invasion by the carotid body neoplasm. In 5/16 dogs, mild lymphadenopathy was noted and this involved the ipsilateral medial retropharyngeal lymph node ($n = 4$), ipsilateral mandibular lymph node ($n = 1$), and ipsilateral cranial cervical lymph node ($n = 1$). In 3/5 dogs, no histopathological examination of the lymph nodes was conducted. In 1/5 dogs, draining chronic hemorrhage and histiocytosis was identified in the enlarged mandibular lymph node seen at imaging, and metastatic disease was seen in two medial retropharyngeal nodes that were mildly enlarged at imaging. In 2/16 dogs, severe ipsilateral medial retropharyngeal lymphadenopathy (up to 3 cm diameter) was identified, with caudal displacement of these nodes, and intense contrast enhancement that was homogeneous in one dog and heterogeneous in the other. Histopathology of these nodes was not obtained. These two dogs with very

marked enlargement of the medial retropharyngeal lymph nodes were also dogs in which invasion in the tympanic bulla ($n = 2$) and brain cavity ($n = 1$) was visible.

In two dogs (a French and an English bulldog), a heart base mass was also identified (on CT in one dog and on echocardiography in another), with a tentative diagnosis of chemodectoma. This was confirmed at necropsy and histopathology in one of these two dogs.

In two dogs, the initial radiology report misidentified the paraganglioma as an enlarged medial retropharyngeal lymph node (one on MRI, one on CT). In these dogs, careful retrospective review of the imaging studies revealed that the corresponding node was compressed and displaced caudally by the carotid body mass, but distinct from it (Fig. 10). In another dog, the initial radiology report misidentified the lesion as a likely thyroid carcinoma. However, retrospective inspection of the precontrast series with appropriate window/level showed that two normally sized and shaped

thyroid glands were present in this dog, with the one ipsilateral to the carotid body tumor mildly displaced caudally (Fig. 10).

Discussion

In this retrospective study, we found that carotid body paragangliomas most commonly had a strong heterogeneous enhancement pattern with both CT and MRI. As this was one of the inclusion criteria, all masses were centered near the carotid body. Masses were typically large at presentation, and often invaded into adjacent structures including the base of the skull, tympanic bulla, and cranial cavity, causing neurologic deficits associated with the nerves in that area such as Horner's syndrome, dysphagia, and facial paralysis. The imaging features observed in our study were consistent with those described in previous canine reports.^{11,14,17,25} Our findings were also similar to the reported CT characteristics of these tumors in people.^{5,8,31} In people, a characteristic "salt-and-pepper" appearance of the tumor is reported on MRI, with hyper- and hypointense foci related to slow-flowing and fast-flowing intravascular blood, and intratumoral hemorrhage.^{5,8,30,31} We did not observe this pattern in the dogs for which MRI was available, but only two dogs in our series received an MRI examination.

Pathologic texts describe carotid body tumors as usually well encapsulated, with the majority of tumors being benign.^{3,4} Malignant tumors tend to show microscopic evidence of tumor cells invasion through the tumor capsule and into adjacent structures.^{3,4} In contrast, we observed a relatively high incidence of invasive tumors in this case series, a feature that was reported in a few canine reports.^{21,22,25} It is possible that this represents a random sample bias toward more malignant tumors, however histopathological differentiation between benign and malignant forms is not possible.^{3,4} The best indicator of malignancy is the presence of metastases,^{3,4} which was not systematically documented in this series.

In previous reports of invasive tumors, we could not find clear mention of erosion into the adjacent bony structures in particular the base of the skull and cranial cavity. In our case series, we observed erosion and invasion into the basilar portion of the skull in 6/16 dogs, similar to what is reported to occur in people. In addition to the six dogs that had definitive invasion into the basilar portion of the skull, three dogs had extension of enhancing tissue in direction of the ipsilateral petro-occipital fissure, following the path of the internal carotid artery. This striking feature could be due to the tumor expanding cranially following the path of the carotid sheath, the cervical deep fascia surrounding the carotid artery and extending from the base of the skull around the internal carotid artery, to the thoracic inlet. In two dogs, neoplastic tissue was invading the tympanic bulla

on the side of the tumor and in one of these two, the tumor extended into the external ear canal. The histopathologic diagnosis was obtained through a biopsy of the portion of the mass that was in the ear canal in one dog and through endoscopic biopsy of the bulla in the other dog. The connection of these lesions with a large invasive mass originating in the area of the common carotid bifurcation was only noted on CT. The invasion into the tympanic bullae and area of the petrous temporal bone in these two dogs could raise the question whether these dogs, and possibly some other tumors invading the basilar skull region, could in fact represent other types of paragangliomas, not originating from the carotid bodies. In people, vagal and jugulotympanic paragangliomas are known to cause mass lesions that are focally invasive in the area of the base of the skull, including the petrous portion of the temporal bone and tympanic bulla.⁵⁻⁸ Also, in people, neurologic deficits such as Horner's syndrome or deficits of the nerves VII (facial paralysis), IX (swallowing disorders), X (hoarseness due to vocal cords function alteration), XI (weakness and atrophy of trapezius and sternocleidomastoid muscles) have been reported more commonly with vagal, tympanic, or jugular paragangliomas.⁸ In all of our dogs though, the bulk of the mass and epicenter of the lesion was located very close to the carotid bifurcation, so it seems less likely that these tumors would have an origin in other paraganglions. Head and neck paragangliomas from other paraganglions than the carotid bodies have been reported in the dog, but are very rare,^{1,9} and there was no description of cross-sectional imaging features for these tumors. It is reported in people that when the paragangliomas are very large, as was the case for all of the invasive lesions in our series, the determination of the exact paraganglion of origin of the tumor is often not possible; their histopathological characteristics do not allow differentiation.^{5,8}

In our case series, the median age at time of diagnosis was consistent with previous reports.^{2,10-12,14-22,24,25} The predominance of brachycephalic dogs and predominance of male dogs was also consistent with previous reports.^{16,17,21,22,26,32,33} The clinical signs at presentation in our group were also similar to those previously reported in the literature.^{1,10-16,18-24} Identification of a mass in the cranial cervical region and dyspnea were common. In dogs with carotid body tumors, dyspnea is often caused by the mass effect exerted by the mass and also the dysregulation of the chemoreceptors of the carotid body.¹⁰ Although not observed in our group, a hoarse voice can be found due to compression of the laryngeal nerves.¹⁰ Three of our dogs had Horner's syndrome at presentation, and this has been reported in a clinical case series.¹⁶ This can be explained by the anatomy of the sympathetic innervation of the eye, with the ganglionic synapse between the first and second neurons (cranial cervical ganglion) being located close to the carotid bifurcation, and the fibers of the third neuron

traveling alongside the internal carotid artery. All three dogs with a Horner's syndrome had a large and invasive mass, including two with invasion into the cranial cavity and two with invasion into the tympanic bulla. Other nerve deficits were observed in our study, including facial nerve paralysis ($n = 2$) and hypoglossal nerve deficits ($n = 1$). In another dog, swallowing disorders were described in the medical history, though not specifically characterized in terms of neurologic deficit; it could be hypothesized that these signs were caused by lesions of the glossopharyngeal (IX) and/or hypoglossal (XII) nerves. Such deficits can be explained by the cross-sectional imaging features we observed, with invasion of the masses into the petro-occipital fissure and/or tympanic bulla in all of these dogs. Invasion into the petro-occipital fissure can cause compression or invasion of the nerves traveling through this structure, including the glossopharyngeal, vagus, and spinal accessory nerves, thereby causing swallowing disorders. The hypoglossal foramen is immediately adjacent and dorsal to the petro-occipital fissure, and therefore invasive lesions in that area can cause hypoglossal nerve dysfunction. Invasion into the tympanic bulla and petrous portion of the temporal bone can cause lesions of the facial nerve and facial nerve paralysis.

Retrospective reviews of the original radiology report in our case series revealed that the lesions were confused with other structures such as an enlarged medial retropharyngeal lymph node or a thyroid gland tumor in three dogs. Confusion of carotid body tumors with thyroid carcinoma on CT has been reported previously,²⁵ and radiologists should be aware of this possibility. Carotid body tumors are located more cranially than thyroid neoplasms, and are closely associated with the carotid bifurcation. Close inspection of the precontrast CT series with appropriate window and level should allow identification of the normal thyroid glands caudal to the mass, due to their iodine content.²⁵ Confusion with enlarged medial retropharyngeal lymph nodes can be caused by the anatomic location of carotid body tumors (adjacent to these lymph nodes, medial to the mandibular salivary glands) with a similar mass effect. Large carotid body tumors may displace the medial retropharyngeal lymph node caudally and compress it, making identification difficult. Careful inspection of multiplanar reformatted series may help the radiologist to identify these structures. The presence of a large mass in that area with no other visible cause that would explain isolated severe medial retropharyngeal lymphadenopathy (e.g., oral cavity neoplasm) should alert the radiologist to consider another etiology such as carotid body tumor.

In people, both MRI and CT are commonly used to diagnose and stage head and neck paragangliomas.^{5,8} Although

some references mention that MRI provides more diagnostic information than CT,⁸ a recent study comparing both techniques reported a higher sensitivity of multidetector CT in differentiating paragangliomas from other similar head and neck lesions.³¹ However, that same study concludes that, due to inherent benefits and disadvantages of both techniques, they should be utilized in conjunction, for accurate diagnosis and grading.³¹

One limitation of this study is the lack of histopathology in 3 of the 16 dogs. We relied on cytological characteristics consistent with a neuroendocrine tumor, together with the anatomic location characteristic of carotid body origin to establish a diagnosis. Also, due to cyto- and histopathological similarities of paragangliomas with other neuroendocrine neoplasias such as medullary thyroid carcinoma or parathyroid carcinomas, immunohistochemistry is necessary for a definitive identification.³⁴⁻³⁶ We considered that the neuroendocrine characteristics seen at cytology or histopathology, together with the location of the lesions at imaging were enough evidence for a diagnosis of carotid body paraganglioma, but the possibility of ectopic thyroid or parathyroid neuroendocrine tumor cannot be entirely excluded. In small animals, thyroid ectopic tissue has been reported in the cranial mediastinum, heart base, inside of the heart, subcutis of the neck, and at the base of the tongue and hyoid apparatus, but not specifically in the location of the carotid bodies.³⁷ Therefore we have reasonable confidence that the lesions described herein truly represented paragangliomas of carotid body origin. In addition, immunohistochemistry is not perfectly accurate either, as several studies have shown that staining pattern is inconsistent with paragangliomas, with decreased staining observed with increasing malignancy.^{13,38} Another limitation of our study is the lack of pathologic confirmation of changes in the lymph nodes that were enlarged on CT or MRI.

In conclusion, findings indicated that canine carotid body tumors have cross-sectional imaging characteristics similar to humans. These tumors most commonly appeared as strongly enhancing masses centered at the carotid bifurcation, and caused displacement and often entrapment of the external carotid artery. Masses were often locally invasive into the surrounding structures. Some extended craniodorsally toward the petro-occipital fissure, with subsequent involvement of the brain cavity and/or tympanic bulla. This area should therefore be included in scans of dogs with neck masses and neurologic deficits. These masses should be differentiated from enlarged medial retropharyngeal lymph nodes or thyroid neoplasia, and veterinary radiologists should consider this neoplasm in their list of differential diagnoses when appropriate.

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